AN IMMERSION WITH REDUCED SOURCE FACTORS AS AN OPEN IMMERSION FOLLOWED BY A CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-factor-reduced-immersion is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 279-285 and proof span 287-303 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement and proof use the author's notation for the closure with the reduced induced subscheme structure; it is preserved unchanged.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The reduced induced closed subscheme structure on the closure of the image, and the factorization of a morphism into a reduced scheme through that closure, cited from the schemes chapter. Cited at proof lines 288-292. Produces the intermediate closed subscheme and the factorization through it.
- The description of an immersion as a morphism which factors through a closed immersion into an open subscheme of the target. Used at proof lines 293-296. Supplies the open subscheme over which the factorization is by a closed immersion.
- Uniqueness of the reduced closed subscheme structure on a closed subset, cited from the schemes chapter. Cited at proof lines 297-299. Identifies the image with the trace of the closure on the open subscheme.

Source reference. Original definition 01J4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2001–2001. Same-chapter reference used in the retained author text at source line(s) 290, 11389. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0356: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4117–4117. Same-chapter reference used in the retained author text at source line(s) 291, 987, 10250, 13640, 14408, 14563. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01J3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2000–2000. Same-chapter reference used in the retained author text at source line(s) 299, 2977. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $h : Z \to X$ be an immersion. If Z is reduced, then we can factor $h = i \circ j$ with $j : Z \to \overline{Z}$ an open immersion and $i : \overline{Z} \to X$ a closed immersion.*

Proof. Let $\overline{Z} \subset X$ be the closure of $h(Z)$ with the reduced induced closed subscheme structure, see Schemes, Definition 01J4. By Schemes, Lemma 0356 the morphism h factors as $h = i \circ j$ with $i : \overline{Z} \to X$ the inclusion morphism and $j : Z \to \overline{Z}$. From the definition of an immersion we see there exists an open subscheme $U \subset X$ such that h factors through a closed immersion into U. Hence $\overline{Z} \cap U$ and $h(Z)$ are reduced closed subschemes of U with the same underlying closed set. Hence by the uniqueness in Schemes, Lemma 01J3 we see that $h(Z) \cong \overline{Z} \cap U$. So j induces an isomorphism of Z with $\overline{Z} \cap U$. In other words j is an open immersion. □